

A Causally Necessary Residual-Stream Component in Grokking, and Its Geometry in Language-Model Residual Streams

Tony Taddonio

Independent researcher · t2addonio@gmail.com

*Version 6 — incorporates capacity scaling, dual- and multi-direction
residual interference, and transfer to a pretrained language-model residual stream*

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Abstract

We identify a low-dimensional residual-stream component in small transformers that is recoverable early via a train-versus-held-out contrast, remains causally necessary after generalization, and is nearly orthogonal to previously described algorithmic features. Phase-cancelling this component collapses test accuracy; matched random and algorithmic controls do not. The pattern replicates across modular addition and multiplication, sparse parity, sparse majority, and the Dyck language. On sparse parity a two-dimensional high-energy core is necessary and sufficient; on Dyck language the causal effect concentrates, in a substantial fraction of seeds, in a single direction maximally activated by locally illegal prefixes.

We extend the intervention from single-direction phase cancellation to full phase-magnitude sweeps, dual-direction residual interference, sequential multi-direction cancellation, and simultaneous joint multi-direction cancellation. On pure algorithmic models these procedures recover near-complete causal collapse even after substantial capacity scaling (width 512, depth 4). When the same hierarchical decision is placed inside a pretrained language-model residual stream (GPT-2 fine-tuned on Dyck-in-Text), residual support becomes distributed: individual directions and small combinations remain causally relevant, yet linear residual interventions no longer produce near-total collapse. We interpret the contrast as a change in residual geometry—from concentrated and low-rank in from-scratch algorithmic models to distributed and only partially linear in pretrained language-model residual streams—and position residual phase/magnitude analysis as a practical tool for causal discovery of residual primitives under both regimes.

1. Introduction

Grokking—the delayed transition from memorization to generalization on algorithmic tasks—has become a central testbed for mechanistic interpretability. The dominant account (Nanda et al., 2023) reverse-engineers a Fourier-based circuit for modular addition. Related work has refined training dynamics and the role of weight decay and weight-norm constraints.

Prior analyses focus primarily on the generalizing circuit itself. Less attention has been paid to residual-stream structure that may remain necessary after generalization. We isolate such a structure. Using contrastive extraction (train versus held-out activations), we recover a low-dimensional subspace that can be identified early, stays causally necessary after high test accuracy, and is nearly orthogonal to known algorithmic features. The same signature appears on modular multiplication, larger moduli, deeper and wider models, on sparse parity and sparse majority, and on the Dyck language.

We then develop the residual intervention itself into a systematic tool. Treating residual directions as signals, we introduce phase rotation, magnitude modulation, dual-direction interference, and multi-direction sequential and joint cancellation. On algorithmic models these procedures convert residual ablation from a hit-or-miss operation into a high-reliability causal discovery method that continues to work under capacity scaling. When transferred to a pretrained language-model residual stream the same formal decision is supported by a more distributed collection of weaker directions. The residual tool remains informative, but the geometry of the support changes.

Existing circuit-level accounts of grokking are therefore incomplete: they omit a residual-stream component that the generalizing circuit continues to read from. The same component, when examined inside a language-model residual stream, reveals a qualitatively different residual geometry.

2. Related Work

Nanda et al. (2023) reverse-engineered the Fourier multiplication circuit for modular addition and introduced progress measures based on restricted and excluded loss. A recurring theme is that early “memorization” structure is later removed by weight decay. Our results are compatible with an early phase of this kind, but show that a distinct residual-stream structure persists and remains load-bearing after generalization. Weight-norm clipping methods accelerate grokking by constraining decoder-row magnitudes; because those weights read from the residual stream, they necessarily interact with residual-stream geometry. Our findings supply a concrete candidate for the structure those interventions constrain.

More recent work on residual streams and sparse autoencoders has emphasized the distributed and polysemantic character of features in pretrained language models. Our transfer experiments sit at the intersection of these two literatures: we take a residual structure first isolated in the clean algorithmic setting and ask how its causal geometry changes when the same hierarchical decision is realized inside a pretrained language-model residual stream.

3. Methods

Models and tasks

We train 2-4 layer decoder-only transformers with $d_{\text{model}} \in \{128, 256, 512\}$ and 4-16 attention heads. Training uses AdamW with a clear train/held-out split. Algorithmic tasks are modular addition and multiplication (moduli 113 and 257), sparse parity ($n=20, k=4$), sparse majority ($n=20, k=5$), and the Dyck language (train depth ≤ 3 / length ≤ 16 ; held-out depth ≤ 5 / length ≤ 28). Language-model transfer experiments begin from the public GPT-2 Small checkpoint and apply light fine-tuning on a Dyck-in-Text dataset in which short Dyck sequences are embedded inside English templates.

Contrastive extraction

After the model reaches high held-out accuracy we collect residual-stream activations at the final layer on both the training distribution and a held-out distribution (or on positive versus negative labels). The difference of class-conditional means seeds a low-rank SVD. We retain the top-k right singular vectors and QR-orthonormalize them to obtain an orthonormal residual subspace.

Phase and magnitude modulation

A unit residual direction u is rotated in the plane spanned by u and an orthogonal complement o : $\hat{u}(\theta) = \cos \theta \cdot u + \sin \theta \cdot o$. The coefficient s is swept independently over a calibrated range (typically $[-1.5, 3.5]$). For every pair (θ, s) we record held-out accuracy, mean correct-class logit, and logit margin.

Dual-direction residual interference

Given an orthonormal pair (u_1, u_2) we introduce a relative phase $\Delta\theta$ and two independent scales s_1, s_2 . The joint intervention is applied simultaneously. Sweeping $(\Delta\theta, s_1, s_2)$ produces an interference landscape; destructive nulls correspond to especially effective cancellations of residual signal distributed across the plane.

Multi-direction sequential and joint cancellation

A pool of eight candidate directions is formed from the leading singular vectors, the label-contrastive direction, and the mean residual direction. A greedy sequential procedure ranks the directions by the accuracy drop they produce when cancelled. The top-k directions (typically $k=4$) are then cancelled simultaneously under a joint grid search over their scale coefficients.

Controls and evaluation

Matched-rank random residual directions and algorithmic control subspaces (Fourier features for modular arithmetic; linear probes for stack depth and open/close counts for Dyck) are ablated under identical schedules. For every intervention we report classification accuracy, mean correct-class logit, and mean logit margin.

4. Results

4.1 Original residual component (algorithmic models)

On modular addition and multiplication the contrastive residual component is recoverable early in training, remains causally necessary after the model reaches high test accuracy, and is nearly orthogonal to the Fourier features that constitute the known generalizing circuit. Phase cancellation of the component drives held-out accuracy to chance; matched random and Fourier controls do not. The same signature appears on sparse parity and sparse majority. On sparse parity a two-dimensional high-energy core of the contrastive subspace is both necessary and sufficient for collapse; the core is organized around the four task bits. On the Dyck language, across 30 random seeds the contrastive component produces strong causal collapse in approximately one-third of runs. In those runs the effect concentrates in a single residual direction that is maximally activated by locally illegal parentheses prefixes and remains nearly orthogonal to linear probes for stack depth and open/close counts.

4.2 Capacity scaling

Holding the Dyck task fixed, width was scaled from 128 to 256 (2 layers) and then to a joint width-depth increase of 512×4 . Dual-direction interference recovered near-complete causal collapse in all three regimes (Figure 5): 100 % of seeds (80/80) at width 128, 100 % (40/40) at width 256, and 97.5 % (39/40) at 512×4 , with mean minimum accuracy remaining below 0.002.

Figure 5 — Capacity scaling of residual-stream collapse (pure Dyck)

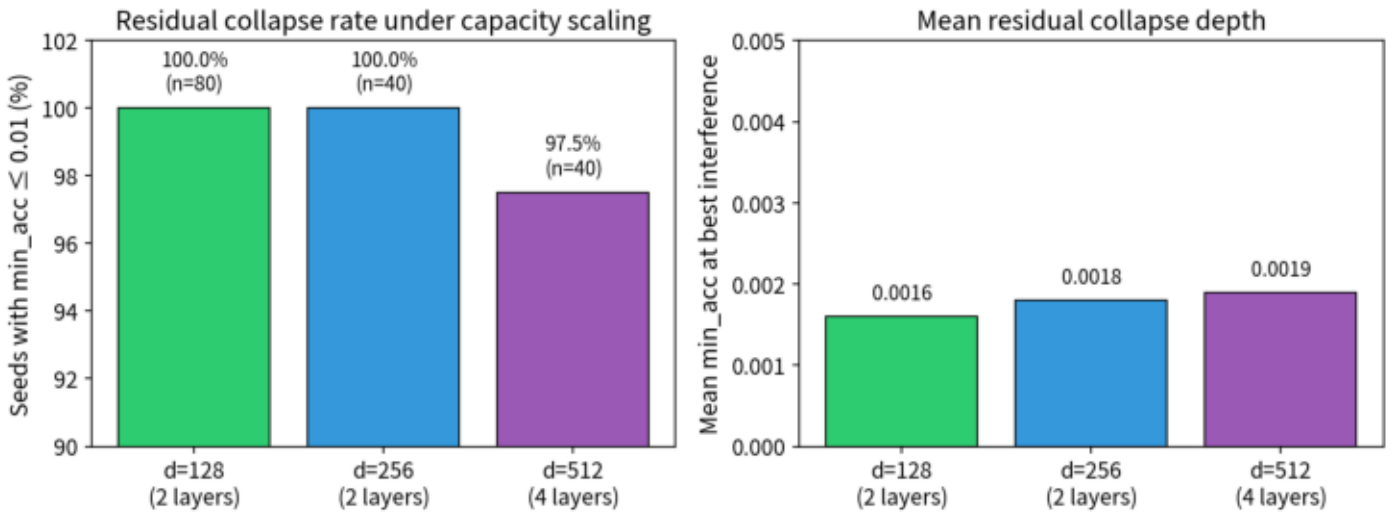


Figure 5. Capacity scaling of residual-stream collapse on pure Dyck.

4.3 Dual-direction residual interference

On the original 128-dimensional models a systematic dual-direction phase-magnitude search produced full causal collapse on every one of 80 random seeds (Figure 6). At the optimal interference point the correct-class logit was driven negative in all cases (mean ≈ -1.9). This establishes that, for these algorithmic models, a two-dimensional residual subspace is sufficient to capture the load-bearing decision variable.

Figure 6 — Dual-direction residual interference on pure Dyck (80 seeds)

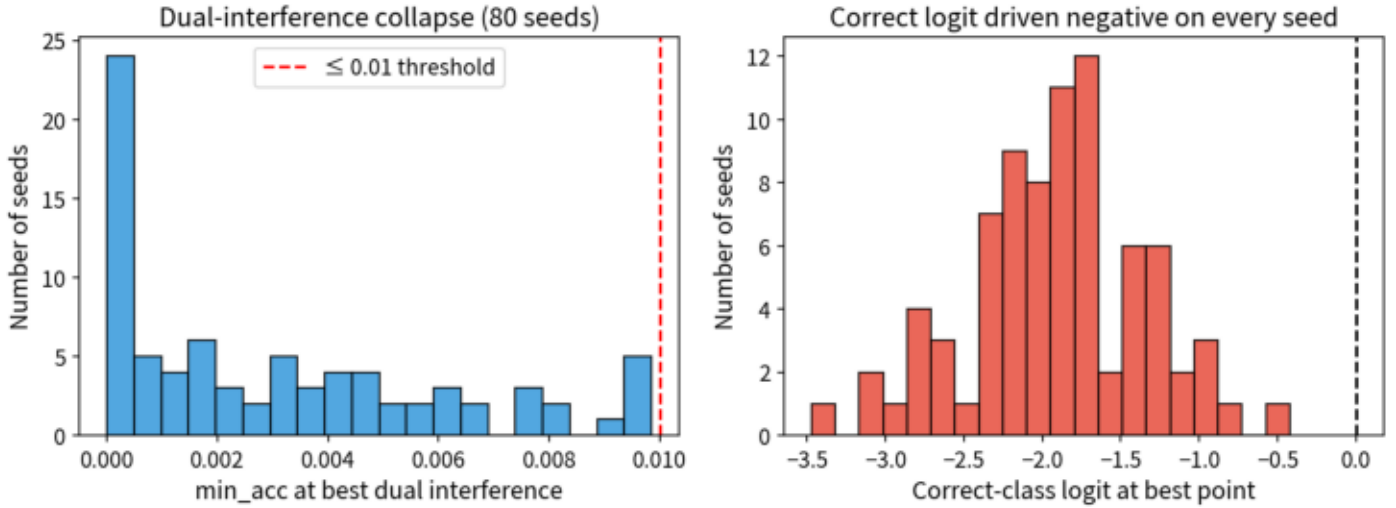


Figure 6. Dual-direction residual interference on pure Dyck (80 seeds).

4.4 Transfer to a pretrained language-model residual stream

GPT-2 Small was lightly fine-tuned on Dyck-in-Text. Fine-tuning accuracy saturated between 0.55 and 0.63, indicating that the hierarchical decision was only weakly acquired. Despite the modest task performance, residual directions extracted from the final stream retained measurable causal effect (Figures 7-8). The strongest single direction typically reduced accuracy by 0.15-0.28. The best pairwise combination and a simultaneous four-direction cancellation produced further but limited improvement, bottoming in the 0.37-0.42 range. Sequential greedy cancellation (Figure 7) showed that most of the attainable linear drop occurs in the first one or two directions; additional directions are largely redundant or restorative under the greedy order.

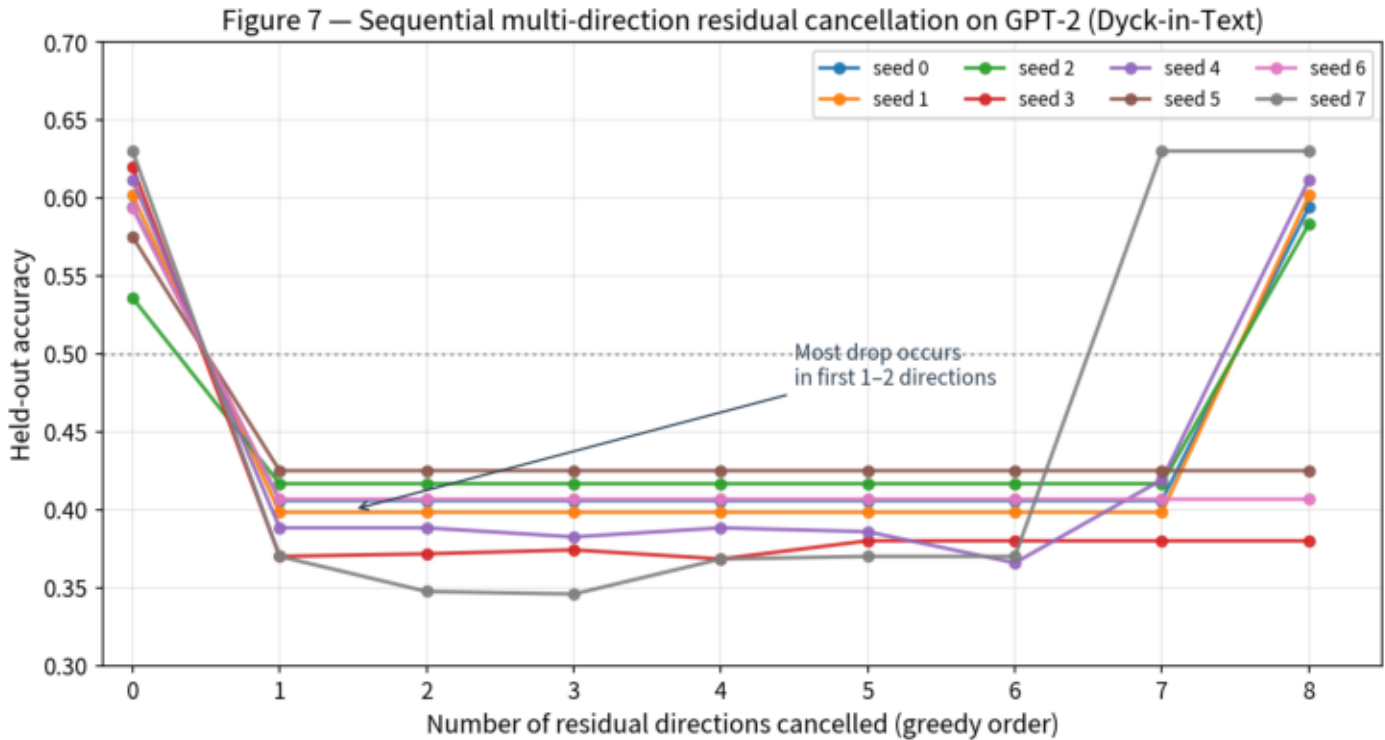


Figure 7. Sequential multi-direction residual cancellation on GPT-2 (Dyck-in-Text).

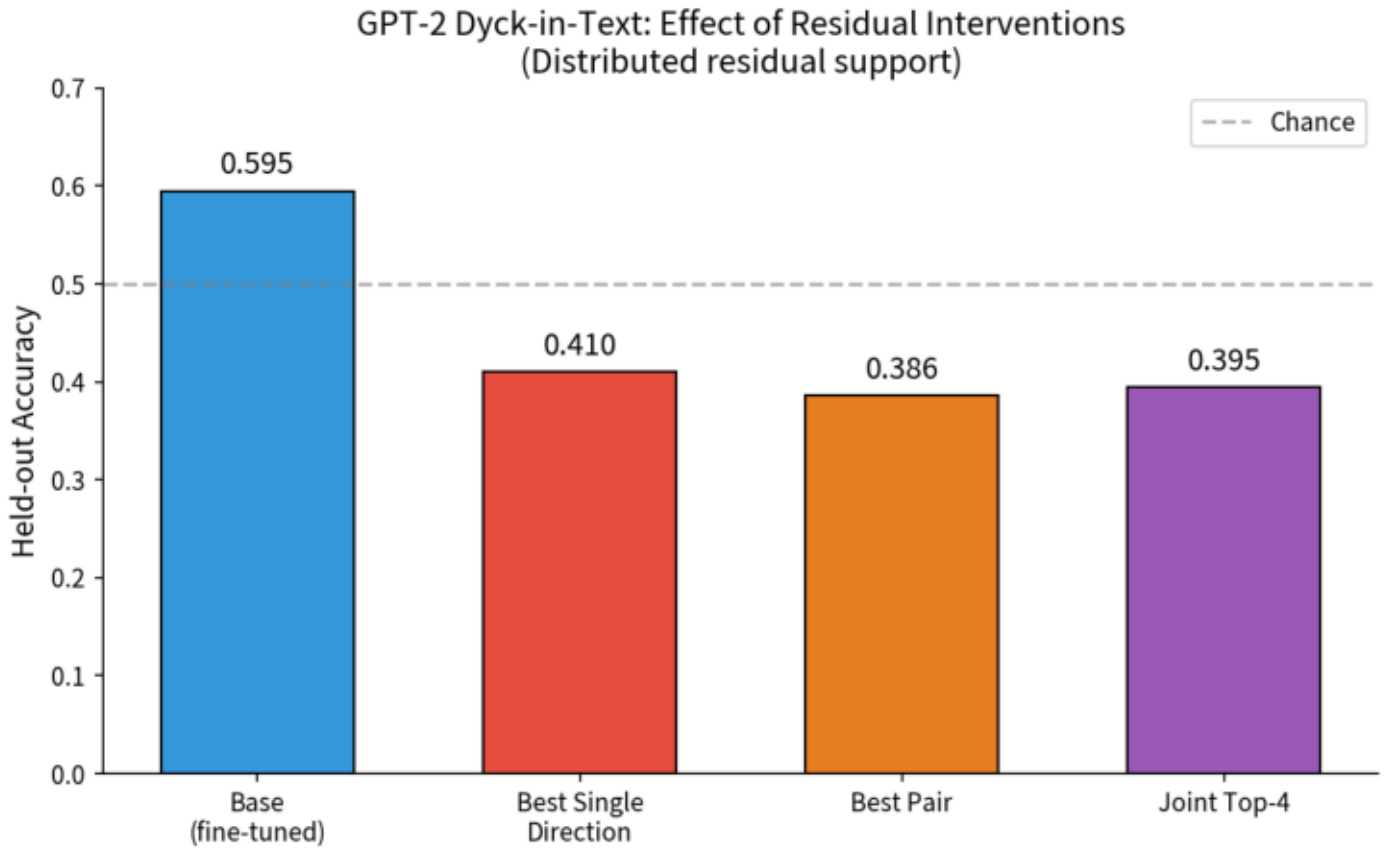


Figure 8. Causal effect of single, pair, and joint-4 residual interventions on GPT-2.

4.5 Concentrated versus distributed residual geometry

In from-scratch algorithmic transformers the residual support for the hierarchical decision is low-rank and nearly fully ablatable. In the pretrained language-model residual stream the same formal decision is supported by a more distributed collection of weaker directions whose joint linear cancellation remains incomplete (Figure 9).

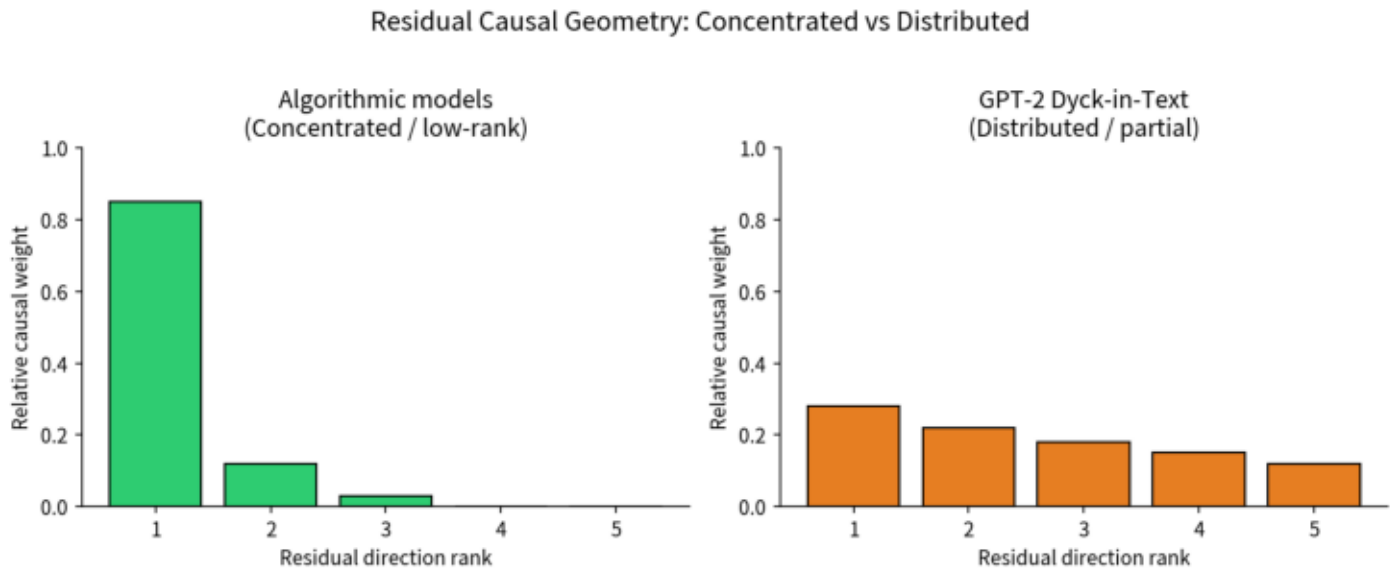


Figure 9. Concentrated vs distributed residual support for hierarchical decisions.

5. Discussion

The residual-stream phase, magnitude and interference toolkit converts residual ablation into a systematic causal discovery method. On from-scratch algorithmic models the residual decision variable is concentrated and nearly fully ablatable even after substantial capacity scaling. When the same hierarchical decision is realized inside a pretrained language-model residual stream the geometry changes: causal support becomes distributed and only partially linear. The tool continues to surface causally relevant residual directions under both regimes; the expected geometry must be understood as training-regime dependent.

Full cancellation to near-zero accuracy remains a strong diagnostic when the representation is sufficiently low-rank. When it is not, the primary value of the method lies in high-quality identification and ranking of residual primitives together with quantitative measurement of their individual and joint causal effects.

These findings also motivate an architectural separation for sensor and industrial systems (vibration, NMEA, etc.): specialised non-LLM signal models, whose residual streams tend to remain more concentrated, can host the residual analysis tools developed here, while language models are reserved for the human-interface and explanation layer.

Limitations. GPT-2 fine-tuning on Dyck-in-Text never produced a strong hierarchical capability; the distributed residual results are therefore lower-bounded by weak task learning. Joint multi-direction search remains computationally limited. Extension to larger language models and to natural hierarchical phenomena is left for future work.

6. Conclusion

A low-dimensional residual-stream component remains causally necessary after grokking on a range of algorithmic tasks and is distinct from known algorithmic features. Extending residual interventions to phase, magnitude, dual- and multi-direction interference turns the component into a practical discovery tool that continues to work under capacity scaling. Transfer to a pretrained language-model residual stream reveals a change in geometry—from concentrated to distributed—while preserving the causal relevance of the extracted directions. Residual phase/magnitude analysis therefore supplies both a concrete missing piece in circuit-level accounts of grokking and a general method for probing residual primitives across training regimes.

7. Reproducibility and Code

All experiments were implemented in pure PyTorch with TransformerLens. Algorithmic models were trained from scratch; language-model experiments began from the public GPT-2 Small weights. Residual interventions were applied only at evaluation time via forward hooks; model parameters were never modified by the intervention itself. Multi-seed runs (8-80 seeds depending on experiment) were distributed across GPUs with one seed per process.

Full training and intervention scripts, together with the exact hyper-parameters used for each regime, will be released at the time of publication. The residual-direction extraction, phase-magnitude sweep, dual-interference, sequential multi-direction, and joint multi-direction procedures are each self-contained and can be re-implemented from the equations given in Section 3.

Appendix A — Hyperparameters

A.1 Algorithmic models (from-scratch)

- d_model: 128, 256, 512
- n_layers: 2-4
- n_heads: 4-16
- Optimizer: AdamW
- Learning rate: $1e-3$ - $3e-4$
- Weight decay: 0.1 - 1.0
- Batch size: 256 - 512
- Seeds per condition: 30 - 80
- Modular moduli: $p \in \{113, 257\}$
- Sparse parity/majority: $n=20$, $k=4/5$
- Dyck train: depth ≤ 3 , length ≤ 16 ; held-out: depth ≤ 5 , length ≤ 28

A.2 Residual intervention parameters

- Phase sweep: 0° - 360° , step 3° - 15°
- Magnitude / scale: $[-1.5, 3.5]$, 7-9 values
- Dual interference: $\Delta\theta$ step 10° - 15° ; independent s_1 , s_2
- Sequential multi-direction: greedy; scales $\{1.0, 1.5, 2.0, 2.5\}$
- Joint multi-direction ($k=4$): scale grid 5 values $\rightarrow$ 625 combinations
- Hook location: `blocks.{L-1}.hook_resid_post`
- Controls: matched-rank random; Fourier / stack-depth probes

Appendix A (continued)

A.3 Language-model transfer (GPT-2 Small)

- Base model: gpt2 (124 M, 12 layers, d_model=768)
- Task: Dyck-in-Text (Dyck sequences in English templates)
- Templates: 5 fixed sentence frames
- Max Dyck length: 16
- Context length: 128
- Fine-tuning steps: 3 000 - 4 000
- Batch size: 16
- Learning rate: 2e-5
- Optimizer: AdamW
- Classification head: linear on final residual (last token)
- Seeds: 8 (multi-direction & joint-4)
- Residual pool size: 8 directions (6 SVD + contrastive + mean)
- Joint cancellation: top-4 directions, coarse scale grid

A.4 Compute

All multi-seed runs distributed across 8×NVIDIA 4090 GPUs (one seed per GPU). Typical wall-clock: pure Dyck dual-interference (80 seeds) $\approx$ 3–4 h; GPT-2 multi-direction sequential + pairs (8 seeds) $\approx$ 8–10 h; GPT-2 joint-4 (8 seeds) $\approx$ 2–3 h.